



Engineering the facilities advanced industry runs on.

EPCM, EPCC and Energy Facility Management for cleanroom-critical industries: semiconductor, data centre, EV battery, pharmaceutical and energy. Established 1994.



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Three parts. Every section is numbered, named and opens with a one-line summary.

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Project References

The work itself, grouped by industry and region.

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The standards every project is held to.

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01

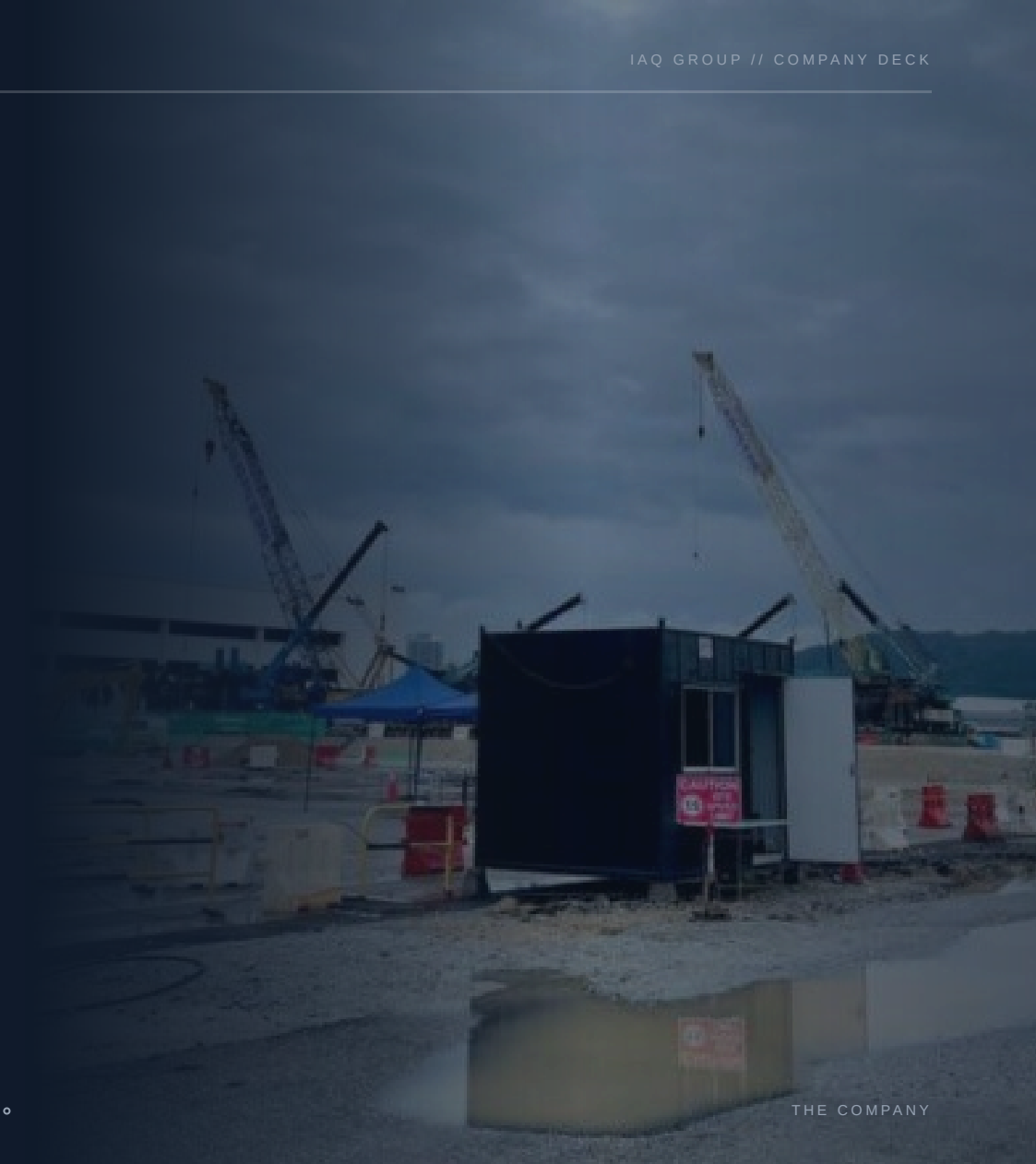
The Company

SECTIONS 01.1 TO 01.8

Established 1994 · 32 years in total facility solutions

450 people across 6 global offices

ISO 9001 · ISO 14001 · ISO 45001 certified



About IAQ

Who we are, what we build, and the models we deliver under.

Established 1994. A trailblazer in total facility solutions.

Engineering, procurement, construction and maintenance for the industries where contamination, uptime and safety decide the outcome, from semiconductor wafer fabs to gigafactories, data centres and pharmaceutical plants.

VISION	To be a regional facility solutions provider with engineering excellence that facilitates technological innovation and advancement in quality of life.
MISSION	Providing innovative, sustainable facility and engineering solutions benefitting our clients and stakeholders, driven by our leadership, employees and partners globally.
DELIVERY MODELS	EPCM · EPCC · ENERGY FACILITY MANAGEMENT

IAQ at a Glance

The company in six numbers, and the certifications that stand behind them.

32 YEARS EXPERIENCE	450 EMPLOYEES	200+ PROJECTS COMPLETED
1.5mil m² CLEANROOM BUILT-UP AREA	6 GLOBAL OFFICES	20t / yr CARBON FOOTPRINT REDUCED

CERTIFICATION & RECOGNITION

INTERTEK ISO 9001:2015	INTERTEK ISO 14001:2015	INTERTEK ISO 45001:2018	CIDB GRADE G7	PKK GRADE G7	HIGHWIRE SAFETY GOLD AWARD	MCIEA 2024 BUILDER OF THE YEAR
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Milestones

Landmark projects from 1994 to 2026, grouped into five eras of growth.

1994 to 2000

Foundation

- 1994

Established as cleanroom specialist
- 2000

First decade of cleanroom delivery in Malaysia

2006 to 2009

Regional Expansion

- 2006

(EPCC) ST Microelectronics, Class 10K cleanroom
- 2008

(PCC) Western Digital PJ, Class 10 cleanroom
- 2009

(GC / D&B) MEMC Ipoh, Class 1 cleanroom

2013 to 2017

Landmark Plants

- 2013

(GC) KLCC DCC Plant, largest DCC plant in Malaysia
- 2016

(PCC) MRT tunnel station M&E · (EPC) Infineon MKZ Class 1K · (EPCC) SilTerra FAB · (PCC) Rapid Siemens powerplant piping · (BOT) Pagoh Edu Hub DCS
- 2017

(GC) Ain Medicare

2020 to 2022

Global Scale

- 2020

(GC) Vital Healthcare
- 2021

(GC / PCC) Robert Bosch · (GC / PCC) P Project
- 2022

(PCC) Microsoft Data Center · (GC / D&B) SilTerra Expansion · (EPCM / D&B) Soitec PR1A Expansion · (GC / EPCM) Infineon IFKM3 Expansion

2023 to 2026

Advanced Technology

- 2023

(GC / D&B) XFAB 40K Expansion
- 2024

(GC / D&B) Progressive Build P Project
- 2025

(GC / D&B) Hasegawa · (PCC) 160MW hyperscale data centre · (EPCC) ESCM Germany · (EPCC) KLCC DCS chiller upgrading
- 2026

(D&B) Tata Dholera DF1 · (EPCM / HU) Micron MSH · (EPCC) GDC Putrajaya chiller upgrading

Global Footprint

Where we operate, and the year we arrived in each market.

From Shah Alam to seven markets.

Offices across Asia Pacific and Europe follow our clients as advanced manufacturing localises. Additional offices operate in Norway, China, France and Poland.

1994

Malaysia

Turnkey design, procurement, construction and commissioning for hi-tech clients. Selangor (HQ), Penang, Johor and Kuching.

2020

Sweden

Gigafactory construction for EV battery manufacture, part of Europe's clean and digital transition.

2023

Singapore

Design & build EPCM services to global semiconductor clients.

2025

Germany

EPCC services following semiconductor expansion in Europe.

2026

India · United States

D&B turnkey for India's first wafer fab, and US entry following localisation of advanced tech facilities.

Core Values

Six commitments that govern how every project is run. Not words, but working rules.



01

Safety First

Our services and works are carried out to the highest standard of safety and ethics.



02

Quality Consistency

We take pride in the quality of solutions delivered, to achieve maximum client satisfaction.



03

Honesty and Integrity

Our core code of conduct, fostering trust, accountability, professionalism and ethical practice.



04

Efficiency & Proficiency

Committed to maximising available resources to achieve the best result collectively.



05

Engineering Capabilities

Our way of working revolves around engineering principles, developing precise and sustainable solutions.



06

Pursuit of Excellence

Devoted to excellence in all our works, providing sustainable solutions to complex challenges.

Business Units

Three companies under one group, covering the full delivery chain.

01

IAQ Solutions Sdn Bhd

ENGINEERING, PROCUREMENT, CONSTRUCTION & COMMISSIONING

EPCC from conception to operation, completion and maintenance. Every stage managed, from initial design through commissioning, so clients in hi-tech industries can bring their visions to life.

02

IAQ Utility Solutions Sdn Bhd

PROCESS CRITICAL UTILITIES & TOTAL TOOL INSTALL

EPCM partner for semiconductor manufacturing: engineering, procurement and construction management of process-critical utility infrastructure and total tool installation, bridging facility readiness and manufacturing start-up to accelerate fab ramp-up.

03

IAQ Energy Facility Management Sdn Bhd

ENERGY MANAGEMENT

Energy management solutions that optimise operations and reduce carbon footprint, keeping facilities running at the highest levels of efficiency and sustainability.

Scope of Services

Twelve packages, self-performed and integrated under one contract.

01 Cleanroom System ISO 3 to ISO 8 · Class 1 to Class 100K	02 Air Conditioning System ACMV, make-up air, recirculation	03 Process Utilities CDA, PCW, PV	04 Process Exhaust System Scrubbed, heat and general exhaust
05 Chemical & Gas Delivery Specialty and bulk gases, chemical delivery	06 Ultra Pure Water UPW generation and distribution	07 Waste Water Treatment Industrial effluent and recovery	08 Fire Protection System Detection, suppression, compliance
09 HT & LV Electrical System Up to 33kV substations	10 Civil, Structural & Architectural Greenfield and brownfield CSA	11 Facility Monitoring & Control FMCS, BMS, PA	12 Tools Hookup Progressive tool install and hookup

Industry Focus

Seven sectors, one discipline: contamination control under code and class.

01 Semiconductor Wafer fab, advanced packaging, test & assembly	02 Data Centre Hyperscale cooling, security, uptime	03 EV Battery Dry rooms, moisture control, explosion proofing	04 District Cooling & Heating Urban energy plant and distribution
05 Photovoltaics Cell and module lines, toxic material handling	06 Pharmaceuticals & Hospitals GMP grades, regulated environments	07 Food & Beverage Hygienic design, quality and safety standards	+ Your facility next Total facility solutions, end to end

PART 02 OF 03

02

Project References

SECTIONS 02.1 TO 02.11

200+ projects completed across three regions

Over 1.5 million m² of cleanroom built

ISO 3 to ISO 8 · Class 1 to Class 100K

Portfolio Overview

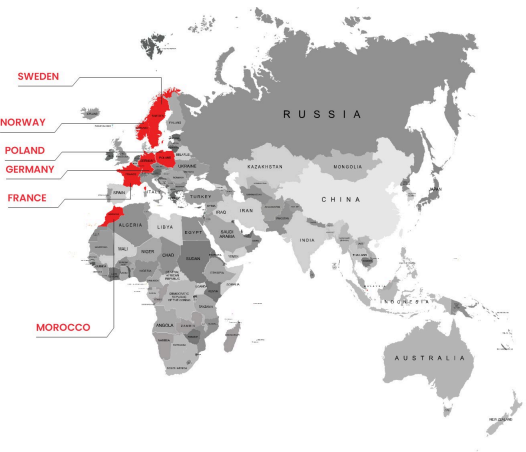
How the reference book breaks down by region, and which sections cover what.



02.2 TO 02.4 · 02.8 TO 02.10

Malaysia & Singapore

Semiconductor, data centre, pharmaceutical, photovoltaics, district cooling and energy.



02.5 TO 02.6

Europe & Morocco

EV battery gigafactories, wafer fabs and advanced semiconductor in Sweden, Norway, France, Poland and Morocco.



02.7

China

Semiconductor, display, pharmaceutical, automotive and R&D facilities.

200+

PROJECTS COMPLETED

1.5 mil m²

CLEANROOM BUILT

5

CONTRACT MODELS

GC · EPCC · EPCM · PCC · Tool Hookup

KULIM, KEDAH · GREENFIELD WAFER FAB

Infineon Kulim Wafer Fab 3

CLEANROOM	ISO 4 · 5 · 6 · 7 (CLASS 10 · 100 · 1K · 10K)
SCOPE	PCC for cleanroom and mechanical works
DESCRIPTION	General contractor for WP06, KLM3 Expansion, 35,000 m ² cleanroom area

ADVANCED PACKAGING · 75,000 M²

P Project

CLEANROOM	ISO 4 TO 7 (CLASS 10 TO 10K)
SCOPE	Design & build GC: CSA, MEP, process utilities, chemical & gases system, waste treatment
DESCRIPTION	Progressive-built EPCM contract model for the client's advanced packaging facility

Semiconductor · Malaysia

Wafer fab, advanced packaging, test and assembly, ISO 3 to ISO 7.



KUCHING, SARAWAK

XFAB Kuching 40K Expansion

CLEANROOM	ISO 5 · 6 · 7 (CLASS 100 · 1K · 10K)
SCOPE	CSA, mechanical, electrical & plumbing, process utilities, UPW, chemical & gases
DESCRIPTION	GC design & build for wafer fab extension, CUB and supporting facilities. Capacity 30k to 40k, 40,000 m ² built-up



GREENFIELD PLANT

SOITEC PR1A Expansion

CLEANROOM	ISO 4 · 5 · 6 · 7 (CLASS 10 · 100 · 1K · 10K)
SCOPE	EPCM design & build for fab, including all related MEP
DESCRIPTION	EPCM consultant for PR1A new greenfield plant, 45,000 m ² built-up

Semiconductor · Malaysia

Continued. Greenfield plants and fab expansions delivered as GC and EPCC.



GREENFIELD · TEST & MANUFACTURING

Robert Bosch Testing Manufacturing Plant

CLEANROOM	ISO 6 · 7 (CLASS 1K · 10K)
SCOPE	Main building CSA, MEP, process utilities and cleanroom package
DESCRIPTION	General contractor for greenfield testing and manufacturing plant, 25,000 m ² built-up



FAB1E A/B · NEW CUB

Wafer Fab Facility & Expansion

CLEANROOM	ISO 5 (CLASS 100)
SCOPE	EPCC new building: CSA, mechanical, electrical, cleanroom, process utilities, specialty and bulk gases, chemical delivery
DESCRIPTION	EPCC and design & build for FAB1E A/B expansion. 1,000 m ² cleanroom, 5,000 m ² new CUB

Semiconductor · Malaysia

Continued. Test, probe and assembly plants for tier-one device makers.



MELAKA

Texas Instruments, Melaka

CLEANROOM	ISO 5 · 6 · 7 (CLASS 100 · 1K · 10K)
SCOPE	Mechanical, electrical, process utilities, testing & commissioning
DESCRIPTION	Facilitation works for residual management, 32,000 m ² built-up



MELAKA · BLOCK 8

Infineon Technologies (M) Sdn Bhd

CLEANROOM	ISO 6 · 7 (CLASS 1K · 10K)
SCOPE	EPCM-GMP for civil & structural, cleanroom, air-conditioning, process utilities and multi-storey carpark
DESCRIPTION	Block 8 testing, probe & assembly plant, 43,000 m ² built-up

Semiconductor · Malaysia

Continued. Class 1 cleanroom capability and multi-storey production works.



IPOH, PERAK · GREENFIELD

MEMC Ipoh Sdn Bhd

CLEANROOM	ISO 3 · 4 · 5 · 6 · 7 (CLASS 1 TO 10K)
SCOPE	Mechanical, electrical, process utilities, cleanroom, fire fighting, FMCS, CSA and tools hookup
DESCRIPTION	Greenfield project. The tightest cleanroom class in the portfolio, ISO 3 (Class 1)



JOHOR

STMicroelectronics Sdn Bhd

CLEANROOM	ISO 7 (CLASS 10K)
SCOPE	Cleanroom architectural, ACMV, electrical, process utility and cleanroom works
DESCRIPTION	Three-storey production factory

Semiconductor · Singapore

Design & build EPCM in Singapore since 2023: photomask, tool hookup and fit-out.



PHOTOMASK

Teksend Photomask

CLEANROOM	CLASS 1 TO CLASS 10K
SCOPE	EPCC for cleanroom, ACMV, electrical, process utility, BMS, PA, WWT, UPW, tool hookup



TOOL HOOKUP

Micron MSH Tool HU

SCOPE	Progressive tool hook up
DESCRIPTION	2,000 progressive tool hookup services for MSH



FIT-OUT

Micron F10A & F10NX

SCOPE	Interior design & mechanical fit-out works
DESCRIPTION	F10NX office optimisation

CLIENT CONFIDENTIAL - 160 MW

Hyperscale Data Centre

SCOPE

Construction, completion, testing & commissioning for mechanical packages 1 & 2: office, data halls, mechanical utility building

ALSO DELIVERED

Microsoft DTC-KUL 03. PCC for CHW and CW piping works, 9.6 MW data centre, KUL03 Phase 1

SKELLEFTEÅ, SWEDEN · 62,000 M²

Northvolt AB Gigafactory

ENVIRONMENT DRY ROOM

SCOPE Cleanroom and dry room architecture works

DESCRIPTION Europe gigafactory for lithium-ion battery manufacture

EV Battery · Europe

Dry rooms for the gigafactories: moisture control and contamination control at scale.



NORWAY · BATTERY COAST

Morrow Batteries

ENVIRONMENT	DRY ROOM
SCOPE	Cleanroom and dry room architectural system, ACMV
DESCRIPTION	8,600 m² built-up



FRANCE · GREENFIELD

ACC Phase 1

ENVIRONMENT	DRY ROOM
SCOPE	Cleanroom, dry room architecture works
DESCRIPTION	8,000 m² dry room system, 35,000 m² built-up



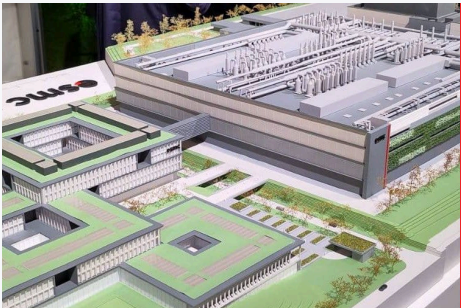
FRANCE

Envision France

ENVIRONMENT	DRY ROOM
SCOPE	Cleanroom, dry room architecture works
DESCRIPTION	Battery manufacturing facility

Semiconductor · Europe & Morocco

Following the fabs into Europe: advanced-node cleanrooms and progressive hookup.



CLIENT CONFIDENTIAL

28/22nm CMOS · 16/12nm FinFET Plant

CLEANROOM	CLASS 1 TO CLASS 10K
SCOPE	Cleanroom & mechanical general contractor



PARIS, FRANCE

Soitec Semiconductor

SCOPE	EPCC for cleanroom and MEP
DESCRIPTION	11,000 m² built-up



CASABLANCA, MOROCCO

STMicroelectronics

CLEANROOM	ISO 5 (CLASS 100)
DESCRIPTION	Cleanroom, MEP and hookup, 60,000 m² built-up



PARIS, FRANCE

Xfab Semiconductor

CLEANROOM	ISO 5 (CLASS 100)
DESCRIPTION	Progressive hookup work, 5,000 m² built-up

Semiconductor & Display · China

EPCC across semiconductor, display and precision manufacturing. 23 further projects in section 02.11.



HANGZHOU · 23,400 M²

Ferrotec Semiconductor

CLEANROOM ISO 4 · 5 (CLASS 10 & 100)

SCOPE EPCC for cleanroom, MEP and hookup works



WUXI · 8,000 M²

Infineon Technologies

CLEANROOM ISO 5 (CLASS 100)

SCOPE EPCC for cleanroom, MEP and hookup works



NINGBO · 22,000 M²

SMIC

CLEANROOM ISO 5 · 6 (CLASS 100 & 1K)

SCOPE EPCC for mechanical, electrical and plumbing works



SUZHOU · 95,000 M²

Maxtor Technology

CLEANROOM ISO 5 · 6 (CLASS 100 & 1K)

SCOPE Cleanroom and MEP works

Photovoltaics

Solar cell and module manufacturing, including toxic material and waste handling.



KULIM, MALAYSIA

First Solar Malaysia

SCOPE	PCC for M&E, KMW building; CSA and M&E for forming gas plant; M&E for SCO1
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MELAKA, MALAYSIA

AUO Sunpower

SCOPE	CSA, utilities, HVAC, electrical, testing & commissioning
DESCRIPTION	Fab 3A solar cell manufacturing



CHINA · 57,000 M²

Chint Solar Technology

CLEANROOM	ISO 8 (CLASS 100K)
SCOPE	Cleanroom, ACMV, process utilities and tools hookup

Pharmaceutical & Medical

GMP grades and regulated environments, from parenteral facilities to medical device plants.



GREENFIELD · MEDICAL DEVICE

Insulet

CLEANROOM	ISO 8 (CLASS 100K)
SCOPE	Design and build mechanical, cleanroom & electrical package



GREENFIELD · MEDICAL DEVICE

Vital Healthcare

CLEANROOM	ISO 8 (CLASS 100K)
SCOPE	GC for CSA, 33kV substation and MEP



PUCHONG, SELANGOR

Pharmaniaga

CLEANROOM	ISO 5 · 7 · 8 (CLASS 100 · 10K · 100K)
SCOPE	Cleanroom system and ACMV for a small volume parenteral facility



PHARMACEUTICAL BUILDING

Ain Medicare

CLEANROOM	ISO 8 (CLASS 100K)
SCOPE	EPCC CSA, cleanroom system and M&E

GAS DISTRICT COOLING (M) SDN BHD

KLCC District Cooling Plant

DESCRIPTION	LARGEST DISTRICT COOLING CENTRE IN MALAYSIA
SCOPE	ACMV, electrical & instrumentation, fire protection, process utility, plumbing & sanitary, water treatment, CSA
ALSO DELIVERED	GDC Putrajaya plant electrification & chiller replacement (PICC) · KLCC DCC centrifugal chillers · Rapid Petronas Pengerang Co-gen, first co-gen plant in SEA · Pagoh Education Hub DCS

Reference Index

A selection from 200+ completed projects. The full reference list is available on request.

MALAYSIA & SINGAPORE

Western Digital (M) Sdn Bhd, Petaling Jaya

WD Media (M) Sdn Bhd, Penang, MA 28 & MA 29

Flextronics Shah Alam Sdn Bhd

Caterpillar Asia, 7T office renovation

M Project, greenfield food flavouring plant, 40,000 m²

P Project, MEP works, mega lab retrofit

National University of Singapore, tissue culture lab

Matrix (M) Sdn Bhd

T Hasegawa, spray dryer & microwave tunnel

Elegant Aura (M) Sdn Bhd

Rapid MCD chemical plant

EUROPE & MOROCCO

Northvolt AB, Skellefteå, 62,000 m²

Morrow Batteries, Norway

ACC Phase 1, France

Envision, France

Soitec Semiconductor, Paris

Xfab Semiconductor, Paris

Nemotek, Rabat, Morocco

STMicroelectronics, Casablanca

Kimoto, Poland

Sumika, Poland

CHINA

All-Cent RF Technology, Wuxi, 25,600 m²

Kunshan Visionox Display, 27,000 m²

GS Magicdrive Inc., 68,000 m²

Shantou Goworld Display

M-Flex (Suzhou), 25,000 m²

MMI Industries (Wuxi)

Huawei Technologies Co., Ltd

Shanghai Roche Pharmaceuticals

Suzhou Ascentage Pharma, 25,800 m²

Jiangsu GenScript Biotech

Mindray Group Co., Ltd

Unilever · GE · Dow · Rohm and Haas R&D centres

PART 03 OF 03

03

Safety, Quality & ESG

SECTIONS 03.1 TO 03.4

Highwire Gold safety award

ISO 45001:2018 occupational health & safety

20 tons per annum carbon footprint reduction

ESG Commitments

Three pillars, applied to operations and to every project we take on.

E

Environmental Responsibility

Minimising environmental footprint through sustainable design and construction. Energy-efficient solutions and eco-friendly technologies, prioritised at design stage.

S

Social Accountability

The well-being and safety of employees and communities first. A diverse, inclusive work environment and ethical labour practices.

G

Corporate Governance

Transparency, accountability and ethical decision-making, with the highest standards of compliance, building trust with clients, partners and stakeholders.

Safety & Recognition

How our standards are independently verified and recognised.

Judged on the things that actually matter.

MCIEA award recipients are evaluated across company performance, project management, technical expertise, innovation, quality, safety and sustainability. The same criteria our clients audit us against.

Builder of the Year 2024

Malaysian Construction Industry Excellence Awards (MCIEA), awarded by CIDB.

Highwire Gold

Gold-rated contractor safety performance.

ISO 45001:2018

Occupational health and safety management, certified by Intertek.

Closing

Your Total Facility Solutions Provider.

We infuse safety, quality, efficiency and excellence into every project.

From engineering design and project management to construction and maintenance, across semiconductor manufacturing, clean energy and every environment in between. IAQ is your trusted partner for total facility solutions.

Contacts

For inquiry and business opportunity.



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